

The One-Quarter Fraction 2^{k-2}

2^{6-2} design: $I=ABCE$ and $I=BCDF$

Complete defining relation $I=ABCE = BCDF = ADEF$

Three words $\{ABCE, BCDF, ADEF\}$, Resolution IV

Alias Structure for the 2^{6-2}_{IV} Design with $I = ABCE = BCDF = ADEF$

$$A = BCE = DEF = ABCDF$$

$$B = ACE = CDF = ABDEF$$

$$C = ABE = BDF = ACDEF$$

$$D = BCF = AEF = ABCDE$$

$$E = ABC = ADF = BCDEF$$

$$F = BCD = ADE = ABCEF$$

$$AB = CE = ACDF = BDEF$$

$$AC = BE = ABDF = CDEF$$

$$AD = EF = BCDE = ABCF$$

$$AE = BC = DF = ABCDEF$$

$$AF = DE = BCEF = ABCD$$

$$BD = CF = ACDE = ABEF$$

$$BF = CD = ACEF = ABDE$$

$$ABD = CDE = ACF = BEF$$

$$ACD = BDE = ABF = CEF$$

Construction of 2^{6-2} Design

■ TABLE 8.9

Construction of the 2^{6-2}_{IV} Design with the Generators $I = ABCE$ and $I = BCDF$

Run	Basic Design				$E = ABC$	$F = BCD$
	A	B	C	D		
1	—	—	—	—	—	—
2	+	—	—	—	+	—
3	—	+	—	—	+	+
4	+	+	—	—	—	+
5	—	—	+	—	+	+
6	+	—	+	—	—	+
7	—	+	+	—	—	—
8	+	+	+	—	+	—
9	—	—	—	+	—	+
10	+	—	—	+	+	+
11	—	+	—	+	+	—
12	+	+	—	+	—	—
13	—	—	+	+	+	—
14	+	—	+	+	—	—
15	—	+	+	+	—	+
16	+	+	+	+	+	+

The 2^{6-2} Design

- Uses of the **alternate** fractions

$$E = \pm ABC, \quad F = \pm BCD$$

- Any subset of the original six variables that is not a word in the complete defining relation will result in a full factorial design
 - Consider $ABCD$ (full factorial)
 - Consider $ABCE$ (replicated half fraction)
 - Consider $ABCF$ (full factorial)

The General 2^{k-p} Fractional Factorial Design

- 2^{k-1} = one-half fraction, 2^{k-2} = one-quarter fraction, 2^{k-3} = one-eighth fraction, ..., $2^{k-p} = 1/2^p$ fraction
- $2^p - 1$ words, each effect has $2^p - 1$ aliases

$$I = \dots = \dots$$

$$A = \dots = \dots$$

$$\vdots$$

- $k-p$ independent factors (columns), p dependent factors
- Design is not unique; we can use the maximum resolution criterion to select the best design

TABLE 8.14**Selected 2^{k-p} Fractional Factorial Designs**

Number of Factors, k	Fraction	Number of Runs	Design Generators
3	2_{III}^{3-1}	4	$C = \pm AB$
4	2_{IV}^{4-1}	8	$D = \pm ABC$
5	2_{V}^{5-1}	16	$E = \pm ABCD$
	2_{III}^{5-2}	8	$D = \pm AB$
			$E = \pm AC$
6	2_{VI}^{6-1}	32	$F = \pm ABCDE$
	2_{IV}^{6-2}	16	$E = \pm ABC$
			$F = \pm BCD$
	2_{III}^{6-3}	8	$D = \pm AB$
			$E = \pm AC$
			$F = \pm BC$

■ TABLE 8.14

Selected 2^{k-p} Fractional Factorial Designs

Number of Factors, k	Fraction	Number of Runs	Design Generators	Number of Factors, k	Fraction	Number of Runs	Design Generators	Number of Factors, k	Fraction	Number of Runs	Design Generators
3	2_{III}^{3-1}	4	$C = \pm AB$		2_{III}^{9-5}	16	$E = \pm ABC$				$L = \pm AC$
4	2_{IV}^{4-1}	8	$D = \pm ABC$				$F = \pm BCD$	12	2_{III}^{12-8}	16	$E = \pm ABC$
5	2_{IV}^{5-1}	16	$E = \pm ABCD$				$G = \pm ACD$				$F = \pm ABD$
	2_{III}^{5-2}	8	$D = \pm AB$				$H = \pm ABD$				$G = \pm ACD$
			$E = \pm AC$				$J = \pm ABCD$				$H = \pm BCD$
6	2_{VI}^{6-1}	32	$F = \pm ABCDE$	10	2_{IV}^{10-3}	128	$H = \pm ABCG$				$J = \pm ABCD$
	2_{IV}^{6-2}	16	$E = \pm ABC$				$J = \pm BCDE$				$K = \pm AB$
			$F = \pm BCD$				$K = \pm ACDF$				$L = \pm AC$
	2_{III}^{6-3}	8	$D = \pm AB$		2_{IV}^{10-4}	64	$G = \pm BCDF$	13	2_{III}^{13-9}	16	$M = \pm AD$
			$E = \pm AC$				$H = \pm ACDF$				$E = \pm ABC$
			$F = \pm BC$				$J = \pm ABDE$				$F = \pm ABD$
7	2_{VII}^{7-1}	64	$G = \pm ABCDEF$				$K = \pm ABCE$				$G = \pm ACD$
	2_{IV}^{7-2}	32	$F = \pm ABCD$		2_{IV}^{10-5}	32	$F = \pm ABCD$				$H = \pm BCD$
			$G = \pm ABDE$				$G = \pm ABCE$				$J = \pm ABCD$
	2_{IV}^{7-3}	16	$E = \pm ABC$				$H = \pm ABDE$				$K = \pm AB$
			$F = \pm BCD$				$J = \pm ACDE$				$L = \pm AC$
			$G = \pm ACD$				$K = \pm BCDE$				$M = \pm AD$
	2_{III}^{7-4}	8	$D = \pm AB$		2_{III}^{10-6}	16	$E = \pm ABC$	14	2_{III}^{14-10}	16	$N = \pm BC$
			$E = \pm AC$				$F = \pm BCD$				$E = \pm ABC$
			$F = \pm BC$				$G = \pm ACD$				$F = \pm ABD$
			$G = \pm ABC$				$H = \pm ABD$				$G = \pm ACD$
8	2_{IV}^{8-2}	64	$G = \pm ABCD$				$J = \pm ABCD$				$H = \pm BCD$
			$H = \pm ABEF$				$K = \pm AB$				$J = \pm ABCD$
	2_{IV}^{8-3}	32	$F = \pm ABC$	11	2_{IV}^{11-5}	64	$G = \pm CDE$				$K = \pm AB$
			$G = \pm ABD$				$H = \pm ABCD$				$L = \pm AC$
			$H = \pm BCDE$				$J = \pm ABF$				$M = \pm AD$
	2_{IV}^{8-4}	16	$E = \pm BCD$				$K = \pm BDEF$				$N = \pm BC$
			$F = \pm ACD$				$L = \pm ADEF$				$O = \pm BD$
			$G = \pm ABC$		2_{IV}^{11-6}	32	$F = \pm ABC$	15	2_{III}^{15-11}	16	$E = \pm ABC$
			$H = \pm ABD$				$G = \pm BCD$				$F = \pm ABD$
9	2_{VI}^{9-2}	128	$H = \pm ACDFG$				$H = \pm CDE$				$G = \pm ACD$
			$J = \pm BCEFG$				$J = \pm ACD$				$H = \pm BCD$
	2_{IV}^{9-3}	64	$G = \pm ABCD$				$K = \pm ADE$				$J = \pm ABCD$
			$H = \pm ACEF$				$L = \pm BDE$				$K = \pm AB$
			$J = \pm CDEF$		2_{III}^{11-7}	16	$E = \pm ABC$				$L = \pm AC$
	2_{IV}^{9-4}	32	$F = \pm BCDE$				$F = \pm BCD$				$M = \pm AD$
			$G = \pm ACDE$				$G = \pm ACD$				$N = \pm BC$
			$H = \pm ABDE$				$H = \pm ABD$				$O = \pm BD$
			$J = \pm ABCE$				$J = \pm ABCD$				$P = \pm CD$
							$K = \pm AB$				

Examples of FFD

EXAMPLE 8.1

Consider the filtration rate experiment in Example 6.2. The original design, shown in Table 6.10, is a single replicate of the 2^4 design. In that example, we found that the main effects A , C , and D and the interactions AC and AD were different from zero. We will now return to this experiment and simulate what would have happened if a

half-fraction of the 2^4 design had been run instead of the full factorial.

We will use the 2^{4-1} design with $I = ABCD$, because this choice of generator will result in a design of the highest possible resolution (IV). To construct the design, we first write down the basic design, which is a 2^3 design, as shown in

■ TABLE 8.3

The 2^{4-1}_{IV} Design with the Defining Relation $I = ABCD$

Run	Basic Design			$D = ABC$	Treatment Combination	Filtration Rate
	A	B	C			
1	–	–	–	–	(1)	45
2	+	–	–	+	ad	100
3	–	+	–	+	bd	45
4	+	+	–	–	ab	65
5	–	–	+	+	cd	75
6	+	–	+	–	ac	60
7	–	+	+	–	bc	80
8	+	+	+	+	$abcd$	96

$$[A] = \frac{1}{4}(-45 + 100 - 45 + 65 - 75 + 60 - 80 + 96) = 19.00 \rightarrow A + BCD$$

$$[AB] = \frac{1}{4}(45 - 100 - 45 + 65 + 75 - 60 - 80 + 96) = -1.00 \rightarrow AB + CD$$

■ **TABLE 8.4**

Estimates of Effects and Aliases from Example 8.1^a

Estimate	Alias Structure
$[A] = 19.00$	$[A] \rightarrow A + BCD$
$[B] = 1.50$	$[B] \rightarrow B + ACD$
$[C] = 14.00$	$[C] \rightarrow C + ABD$
$[D] = 16.50$	$[D] \rightarrow D + ABC$
$[AB] = -1.00$	$[AB] \rightarrow AB + CD$
$[AC] = -18.50$	$[AC] \rightarrow AC + BD$
$[AD] = 19.00$	$[AD] \rightarrow AD + BC$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4 + \hat{\beta}_{13} x_1 x_3 + \hat{\beta}_{14} x_1 x_4$$

where x_1 , x_3 , and x_4 are coded variables ($-1 \leq x_i \leq +1$) that represent A , C , and D , and the $\hat{\beta}$'s are regression coefficients that can be obtained from the effect estimates as we did previously. Therefore, the prediction equation is

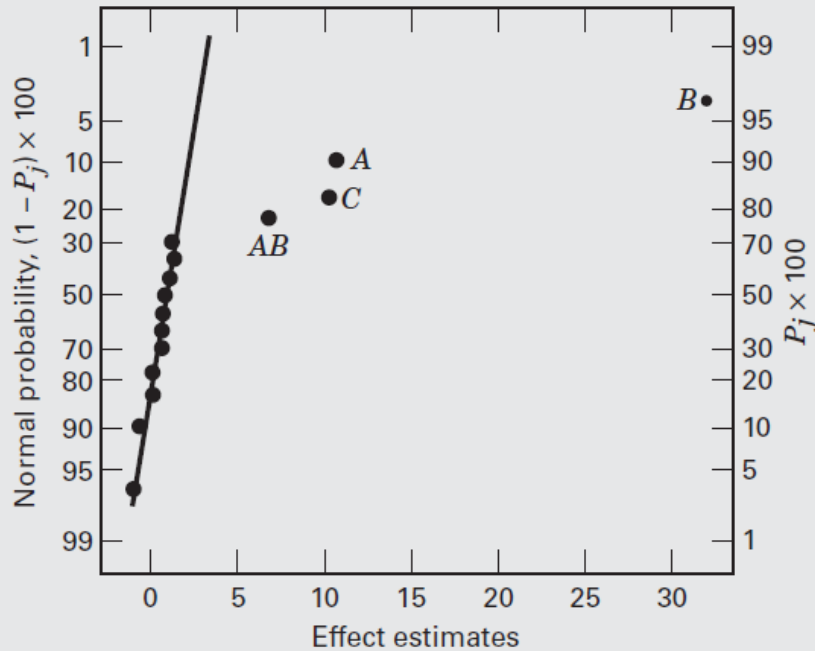
$$\begin{aligned} \hat{y} = & 70.75 + \left(\frac{19.00}{2} \right) x_1 + \left(\frac{14.00}{2} \right) x_3 + \left(\frac{16.50}{2} \right) x_4 \\ & + \left(\frac{-18.50}{2} \right) x_1 x_3 + \left(\frac{19.00}{2} \right) x_1 x_4 \end{aligned}$$

Another Example

■ TABLE 8.5

A 2^{5-1} Design for Example 8.2

Run	Basic Design				$E = ABCD$	Treatment Combination	Yield
	A	B	C	D			
1	–	–	–	–	+	<i>e</i>	8
2	+	–	–	–	–	<i>a</i>	9
3	–	+	–	–	–	<i>b</i>	34
4	+	+	–	–	+	<i>abe</i>	52
5	–	–	+	–	–	<i>c</i>	16
6	+	–	+	–	+	<i>ace</i>	22
7	–	+	+	–	+	<i>bce</i>	45
8	+	+	+	–	–	<i>abc</i>	60
9	–	–	–	+	–	<i>d</i>	6
10	+	–	–	+	+	<i>ade</i>	10
11	–	+	–	+	+	<i>bde</i>	30
12	+	+	–	+	–	<i>abd</i>	50
13	–	–	+	+	+	<i>cde</i>	15
14	+	–	+	+	–	<i>acd</i>	21
15	–	+	+	+	–	<i>bcd</i>	44
16	+	+	+	+	+	<i>abcde</i>	63



■ **TABLE 8.7**

Analysis of Variance for Example 8.2

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-Value
A (Aperture)	495.0625	1	495.0625	193.20	<0.0001
B (Exposure time)	4590.0625	1	4590.0625	1791.24	<0.0001
C (Develop time)	473.0625	1	473.0625	184.61	<0.0001
AB	189.0625	1	189.0625	73.78	<0.0001
Error	28.1875	11	2.5625		
Total	5775.4375	15			